Who drives revenue, and who comes back?

Customer revenue and retention analysis of a UK online retailer, Dec 2009 – Dec 2011

1,033,036 transactions · 5,852 customers · Python, Snowflake, Tableau

Aditya Ranjan · Portfolio project · October 2026

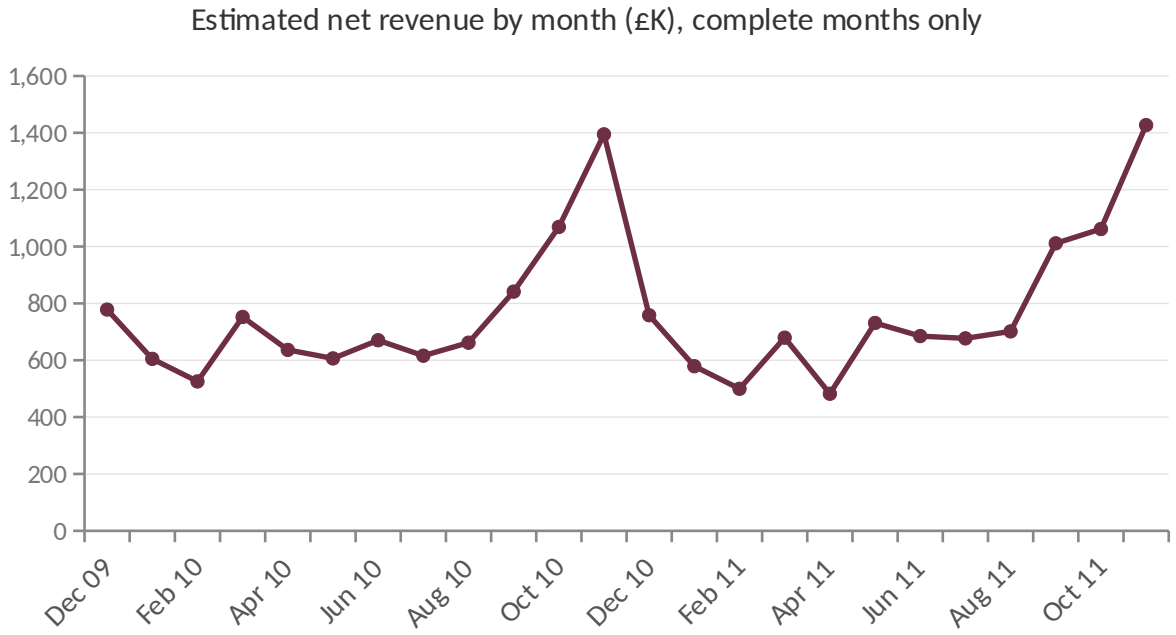
Revenue builds every autumn and peaks in November

£19.60M
gross merchandise sales

£18.89M
estimated net revenue

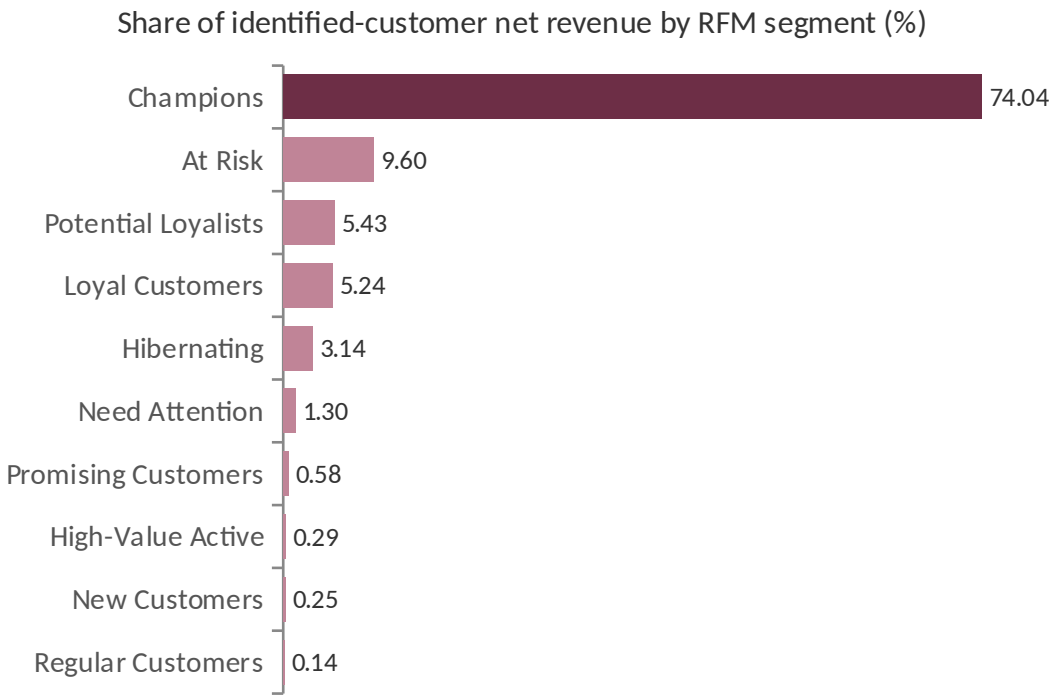
3.65%
cancelled (£716K)

£496
average order value



Nov 2010: £1.39M · Nov 2011: £1.43M, about 1.8–1.9× an average month. Dec 2011 is partial and left out.

A quarter of customers bring three-quarters of revenue



1,448

Champions: 24.74% of identified customers

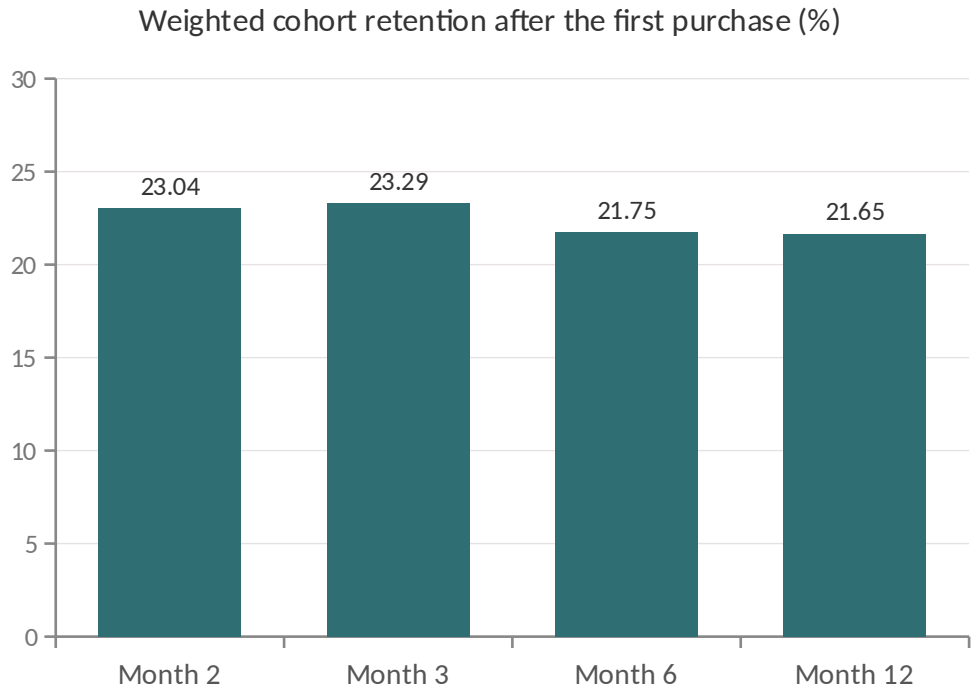
74.04%

of identified-customer net revenue

£8,346

average per Champion, vs £2,789 for all identified customers

Customers who leave mostly leave after the first order



27.65%

of identified customers never bought again

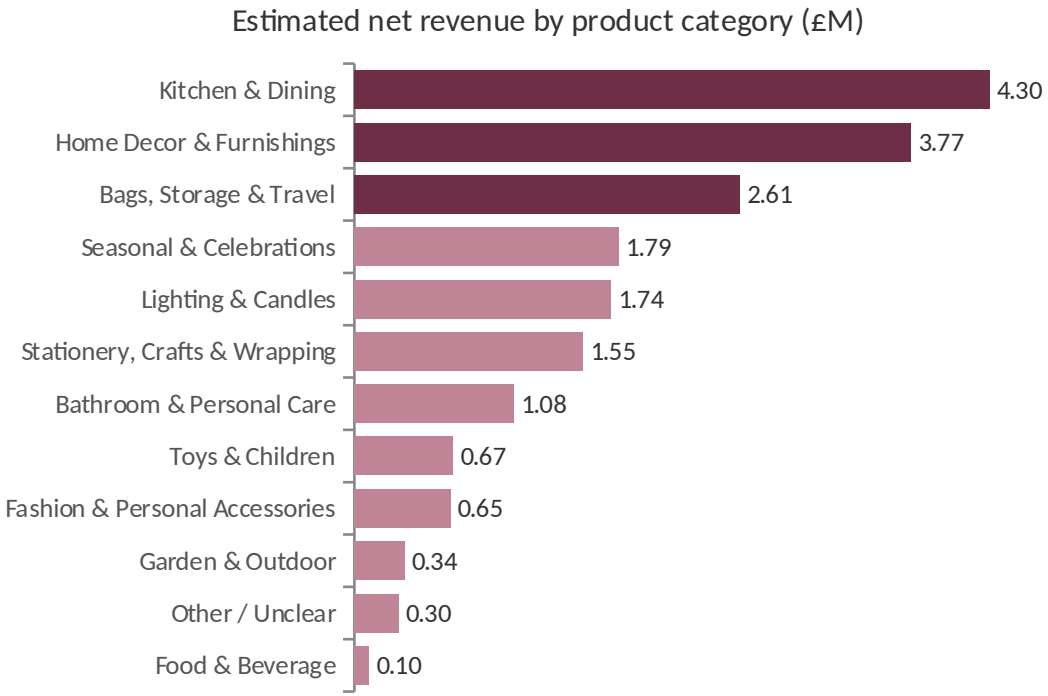
72.35%

came back at least once

96.74%

of identified-customer sales came from those repeat customers

Three categories earn more than half of net revenue



56.5%

of net revenue from Kitchen, Home Decor and Bags

6% → 14%

Bags: share of products vs share of revenue

11% → 3%

Fashion accessories: products vs revenue

Every number was reconciled before reaching a dashboard

1,067,371
raw rows from two overlapping Excel sheets

−34,335
duplicates removed (22,202 in both sheets)

1,033,036
clean rows, loaded to Snowflake RAW → STAGING → ANALYTICS

0.05 pp
largest gap between Tableau and Snowflake segment shares

Flagged, not deleted: a suspected £38,970 entry error was kept and marked for review.

Totals matched at every layer: row counts, revenue and 5,852 customers in RAW, STAGING and ANALYTICS.

Categories checked: 4,725 products classified by a local LLM, with rules and schema checks; none missing.

What I would test next

1
Protect Champions

Test loyalty benefits and account outreach for the 1,448 Champions against broad campaigns. They hold 74% of revenue.

2
Win the second order

Test a follow-up in the first 30 days. 27.65% never return, and most of the drop happens right after the first order.

3
Record why orders are cancelled

Capture a reason code at cancellation. At 3.65%, each percentage point is about £196K of gross sales.

Limits: a public dataset (UCI Online Retail II), so these are associations, not proof of cause, and no business acted on them. Code, SQL and dashboards: github.com/beingbrute/Customer-Revenue-Retention-Analytics